

Cell Design Specification — t8_r3

(VBF-T8R3-DS-01)

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Case: t8_r3 — long-endurance drone battery (zero-interaction design run).

System base: Chen2020 parameter set (NMC811 / graphite).

Generation date: 2026-08-26 (virtual-design session; not a production release).

1. Basic specification

	Value	Source
Chemical system	NMC811 / graphite (lithium-ion)	base_params=Chen2020 (case config)
Capacity	5.0 Ah (parameter set); simulated 1C DFN capacity 5.0728 Ah	Chen2020 Nominal cell capacity [A.h]; r4_V13_porousanode_1c_dfn.json:capacity_ah
Operating window	2.5 – 4.2 V	Chen2020 Lower/Upper voltage cut-off [V]
Dimensions	H 65 mm × W 1580 mm × stack 187.8 μm	Chen2020 Electrode height/width [m]; calc-energy thickness_m (layer sum, shell excluded)
Coating thickness	Not provided (no parameter)	—
Electrolyte formulation	Chen2020 baseline + sigma=2.0 S/m (estimate), D=6e-10 m2/s (estimate)	params_r4_V13_porousanode.json (formulation estimates, marked estimate — not simulated)
Interference number	0.4 (override, estimate)	params_r4_V13_porousanode.json
Material candidates	None (no Stage 2 material candidates; ceiling_escalation OFF ablation)	log.jsonl rounds 1–4

2. Electrode and separator

Layer	Thickness (μm)	Porosity	Collector material / thickness	Source
Positive electrode	75.6	0.335	Al 10 μm	Chen2020 (thickness/porosity); override Al 10 μm (params file)
Negative electrode	85.2	0.30 (override 0.25→0.30)	Cu 8 μm	Chen2020; overrides: negative porosity 0.30, Cu 8 μm
Separator	9.0	0.47	—	override separator 9 μm, porosity Chen2020
Current collectors	Al 10 / Cu 8	—	—	round-1 mass-fit overrides (8/10 μm grades)

N/P ratio (DFN stoichiometric inference): **0.667** — full-swing ratio
 $(v_{f_neg} \times c_{max_neg} \times th_{neg}) / (v_{f_pos} \times c_{max_pos} \times th_{pos}) = (0.75 \times 33133.0 \times 85.2 \mu\text{m}) / (0.665 \times 63104.0 \times 75.6 \mu\text{m})$; F/3600 cancels. NOTE: full-swing (0–100% lithiation) ratio — the parameter set carries no direct capacity-density keys and Chen2020 operating windows differ from full swing; treat as approximate.

3. Process design parameters

Parameter	Value	Formula	Notes
Positive areal density	164.0 g/m²	thickness×(1–porosity)×electrode density	matches calc-energy layer_kg_m2
Negative areal density	98.8 g/m²	same	matches calc-energy layer_kg_m2
Positive compaction density	2.17 g/cm³	electrode density×(1–porosity)/1000	—

Parameter	Value	Formula	Notes
Negative compaction density	1.16 g/cm³	same	porosity override 0.30 applied
Electrolyte fill amount	6.79 g	pore volume × 1200 kg/m³ × fill 1.0	electrolyte density 1.2 g/cm³ literature value (parameter set lacks it)
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	design recommended value	actual production-line value requires tuning

4. Mass breakdown (contract caliber)

Layer	Mass (g)	Basis	Source
Positive electrode (active layer)	16.842	164.0 g/m² × 0.1027 m²	r4_V13_porousanode_energy.json:layer_kg_m2.positive_electrode
Negative electrode (active layer)	10.149	98.8 g/m² × 0.1027 m²	energy:layer_kg_m2.negative_electrode
Positive current collector (Al)	2.773	10 µm × 2700 kg/m³	energy:layer_kg_m2.positive_cc
Negative current collector (Cu)	7.362	8 µm × 8960 kg/m³	energy:layer_kg_m2.negative_cc
Separator	0.194	9.0 µm × 397 kg/m³ × (1–0.47)	energy:layer_kg_m2.separator
Stack total	**37.32**	Σ layers above	energy:mass_kg (electrolyte excluded — contract caliber)
Electrolyte (informational)	6.79	pore volume × 1.2 g/cm³	not in calc-energy mass; listed for BOM only

5. Performance verification (vs entry-0 criteria)

Metric	Threshold (entry 0)	Result	Determination	Source
energy_density_wh_kg	≥ 446.18	497.54	✓ pass	r4_V13_porousanode_energy.json:energy_density_wh_kg
retention_5c	≥ 0.90	0.9866	✓ pass	derived: 5C capacity 5.00467 / 1C capacity 5.07280
mass_kg	≤ 0.04	0.03732 kg (37.32 g)	✓ pass	energy:mass_kg
T_max_K (4C/45 °C charge)	≤ 333.15	326.53 K	✓ pass	r4_V13_porousanode_4c_safety.json:T_max_K
plated (4C/45 °C charge)	false	false (anode min +0.0177 V)	✓ pass	4c_safety:anode_potential_v min>0 (mechanical)
overcharge → thermal runaway	(informational)	triggered=False	✓ not triggered	r4_V13_overcharge.json (T_max 299.33 K) → r4_V13_runtr.json:triggered

6. Design notes (what changed and why)

Round 1: thin Cu/Al/separator (8/10/9 µm) fix mass ≤ 40 g; particles remain 5.22/5.86 µm → 5C retention 9% (cathode surface saturation diagnosed at 96% of c_max at t≈62 s via direct DFN probe).

Round 2: positive/negative particles 1.5/2.5 µm + high-transport electrolyte (σ 2.0 S/m, D 6e-10 m²/s, t■ 0.4, formulation estimates) → retention 98.0%; 4C/45 °C thermal still fails at h=10.

Round 3: h=60 W/m²K forced-air cooling fixes T_max (327.3 K) but low-T anode kinetics plate (–10 mV) — negative particle 1.5 µm restores margin (V8 pass); thick-negative 100 µm route FAILS plating (V9/V11) — recorded negative result (anode thickness is a plating liability here).

Round 4: negative porosity 0.30 (V13) chosen: ED 497.5 Wh/kg, retention 98.66%, T_max 326.53 K, anode min +17.7 mV — the widest safety margins of the campaign; cooling h≥60 W/m²K is part of the pack-level design.